EGX360 - Intelligence Stock Market Analysis and Forecasting Platform

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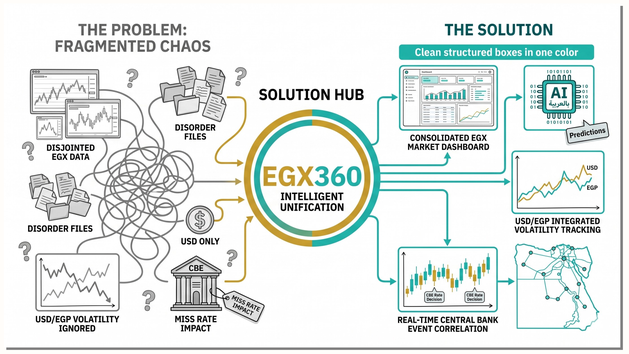
Aims & Rationale

The Problem: Fragmented EGX data lacks unified, real-time AI analysis.

The Goal: EGX360: High-accuracy forecasting using a Stacking Ensemble

Rationale: Cloud-native architecture ensures scalable, high-performance processing.



Summary of Work

Cross-Platform UI Adaptive Flutter dashboard for Mobile and Desktop.

AI Core Stacking Ensemble (XGBoost + LightGBM) for trend forecasting.

Data Infrastructure Automated AWS pipeline for real-time data ingestion.

Backend Services Supabase integration for secure, high-performance management.

NLP Pipeline Real-time sentiment extraction via Hugging Face.

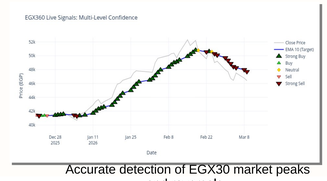
Results & Performance

Model Accuracy	Backtest Alpha	Down Recall
90%	+159.09%	80%

Financial Backtest

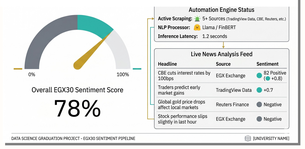
Strategy: 483.42% vs. Market: 324.34%

Live Signals



Accurate detection of EGX30 market peaks and reversals

Automated NLP Engine



Automated News Sentiment Pipeline providing real-time alpha signals

Comparative Analysis

Study	Journal / Rank	Model Architecture	Market Context	Accuracy
Livieris et al. (2025)	Springer (Q1)	SVM, LSTM, GB	55 Global Markets	51%
Park et al. (2024)	MDPI (Q2)	LSTM vs GRU	NASDAQ	65%
Chaudhari et al. (2025)	ScienceDirect (Q2)	Ensemble + Sentiment	BSE/NSE India	72%
EGX360 (This Work)	KSIU Senior Project	Stacking (XGB + LGBM)	EGX30 (Egypt)	90%

System Architecture

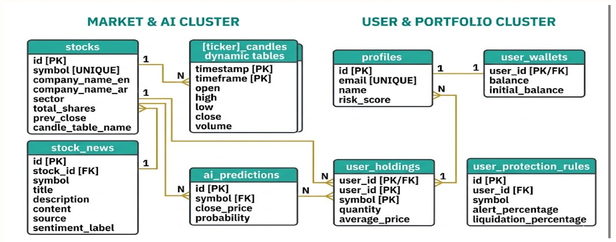


Figure 4.2: Relational Database Schema designed for high-scalability, featuring dynamic candle storage and isolated user portfolio management.

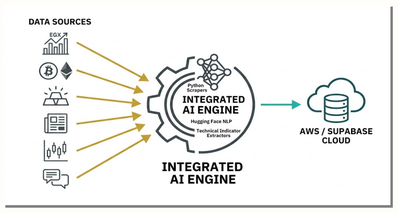


Figure 4.1: Automated Data Ingestion Pipeline utilizing Python scrapers and Hugging Face NLP for real-time sentiment extraction from global financial news.

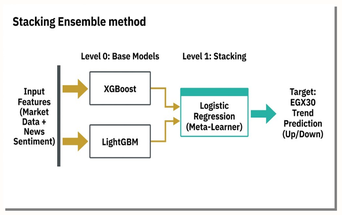


Figure 4.3: Proposed Stacking Ensemble Architecture merging XGBoost and LightGBM with a Logistic Regression Meta-Learner for optimized trend prediction.

Lessons Learned

Technical Expertise

- **Stacking**: Optimized XGBoost & LightGBM base learners.
- **Optuna**: Bayesian optimization for robust model stability.
- **Quant**: Hybrid technical and macroeconomic modeling.
- **Cloud**: Scalable deployment on AWS & Supabase.

Core Knowledge

- **Target Engineering**: EMA10 noise filtering mechanics.
- **Quant Finance**: Gold, USD, and CBE rate correlations.
- **NLP**: Real-time Hugging Face sentiment pipelines.
- **Risk**: Designing automated stop-loss protection rules.

Problem Solving

- **Bias**: Strict chronological splitting for temporal integrity.
- **Noise**: EMA-based smoothing for clean signal extraction.
- **Scale**: Aligning a massive 28-year historical dataset.
- **Latency**: AWS Lambda optimization for real-time inference

Future Work

Live Brokerage Integration: Migrating from paper-trading to real-world execution via local brokerage APIs for automated trade execution.

Multi-Index Expansion: Adapting the Stacking Ensemble model to forecast broader indices like EGX70 and EGX100 for diversified stock analysis.

SHAP-Based Explainability: Utilizing Explainable AI (XAI) to show users exactly how macroeconomic factors (like Gold or CBE rates) influenced each signal.

Conclusion

EGX360 achieves 90% accuracy by synergizing Egypt-specific macroeconomic indicators with a robust Stacking Ensemble. This project bridges the gap between raw data and actionable intelligence, providing retail investors with a production-ready tool for market volatility.